

# **New Evidence on Linear and Non-linear Relation between Returns, Volatility and Trading Volume**

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## **Abstract**

This article examines the linear and non-linear relations between returns (volatility) and trading volume for Indian currency futures contracts. We report a positive contemporaneous relation between futures returns and trading volume which persists even after controlling for heteroskedasticity providing support to mixture of distribution hypothesis (MDH). The results of linear Granger causality show a unidirectional causality from futures returns to volume rejecting the sequential arrival of information (SAI) hypothesis. However, volume does have significant predictive power for futures returns when a non-linear Granger causality framework is used. The results of both linear and non-linear Granger causality between futures returns volatility and trading volume indicate a significant bidirectional relation between the two variables lending support to SAI. These results are robust to alternative measures of trading volume across sub-periods. Further, the results are insensitive to divergence of opinions as proxied by open interest. Overall, we suggest that the short-to medium-term movements in the currency markets are dominated by market microstructure and by fundamentals.

*Key Words:* Trading Volume, Returns, Currency Futures, Volatility, Generalized Method of Moments, Granger Causality.

*JEL Classification:* G12, G13, G14

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## 1. INTRODUCTION

Many of the fundamental variables such as interest rates differentials, money supplies and balance of payments were the primary determinants of exchange rates under the fixed exchange rates regime (Mougoué and Aggarwal, 2011). However, none of these fundamentals are able to explain the increased variability in the exchange rates in the floating periods at least in the short– or medium–run (Frankel and Rose, 1994). Therefore, to account for the lack of explanations, other alternative models are required which would be doing well in modeling the changes in exchange rates. One such potential area of research is the market microstructure models where trading volume, among other variables, better explains the movements in exchange rates (Lyons, 2001). The price–volume relation contains information about market structures (Karpoff, 1987) and helps decide upon the success or failure of futures contracts.

While the relation between price and volume in equity markets has been explored in large body of literature<sup>2</sup>, currency markets have received very little attention (Mougoué and Aggarwal, 2011). Although, in recent times, some papers have started examining the price–volume relation in foreign exchange markets (e.g., see, Melvin and Yin, 2000; Chang, 2005; and Bauwens, Omrane and Giot, 2005); majority of them focus on currency forward markets for developed countries. Very less number of studies are conducted for currency futures markets (e.g., see, Chang, 2005 and Mougoué and Aggarwal, 2011) and none for the emerging markets. We provide a pioneer study on the relation between returns (volatility) and trading volume for currency futures market in an emerging market (India).

We chose currency futures markets as they offer several advantages over forward markets. First, unlike forward markets, price–volume relation can be examined accurately in currency futures markets, as the contracts are traded on organized exchanges, which provide reliable measures of transaction prices and trading volume. On the contrary, trading volume in the forward market is generated by interbank transactions which could be unreliable. Second, currency futures markets, unlike forward markets, provide important information about open interest which is used as a proxy for heterogeneity of beliefs and a major determinant of trading volume (Bessembinder et al., 1996). We consider the currency pairs USD–INR, EUR–INR, GBP–INR and JPY–INR futures contracts traded on the National Stock Exchange of India Limited (NSE). The NSE

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<sup>2</sup> See for instance, Lee and Rui (2002), Fleming and Kirby (2011), Chen (2012), Chuang, Liu and Susmel (2012), Gebka and Wohar (2013) to mention a few.

introduced the currency futures trading in India only in August, 2008 and, therefore, the market is still very much unexplored. To the best of our knowledge, there is no study that investigates the forecast ability of trading volume to futures returns and their volatility in an emerging currency futures market. Given that currency futures market is one of the largest markets in the world, and Indian rupee has seen wide fluctuations in the recent years, it seems exciting to explore the price–volume relation in the Indian currency futures market.

Much of the empirical evidence on price–volume relation primarily focuses on contemporaneous relation using correlations; few studies explore a dynamic or causal relation (Gallant, Rossi and Tauchen, 1992 and Lee and Rui, 2002). Our contribution to the literature is many fold. *One*, we examine the cotemporaneous and dynamic returns–volume relation using the generalized method of moments (GMM) and Granger causality frameworks respectively. We include the volatility of daily returns as well since the dynamic relation between returns and volume may be impacted by volatility (Lee and Rui, 2002). Specifically, this paper intends to investigate which of the two major theories, mixture of distribution hypothesis (MDH) or sequential arrival of information (SAI), better explains the returns (volatility) and volume relation in the Indian currency futures market<sup>3</sup>. *Two*, in contrast to previous studies, we incorporate two different specifications of trading volume in terms of detrended and unexpected volume to provide robustness check for the results.

*Three*, the next important contribution of this paper is to provide new evidence on the non–linear relation between futures returns (volatility) and trading volume and compare the results with those reported for the developed markets. The traditional linear causality tests may not be suitable to capture the non–linear relation between returns (volatility) and volume (Baek and Brock, 1992; and Hiemstra and Jones, 1994). Examining the non–linear causal relations between futures returns and volume is warranted on the account that the time series of futures prices and volumes are likely to be generated by non–linear processes. We investigate non–linear relation between futures returns (volatility) and trading volume for the four currency futures contracts using a modified version of Baek and Brock (1992) test. Similar to Mougoué and Aggarwal

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<sup>3</sup>Most of the research on the price–volume relation is concerned about mixture of distribution hypothesis (MDH) which states that returns and trading volume are positively related (Harris, 1986; Karpoff, 1987; Ciner, 2002; and Chen, 2012). Second theory, the sequential arrival of information (SAI), states that past prices and volumes can predict each other as the market participants react to new information sequentially (Copeland, 1976; Chen and Daigler, 2008 and Mougoué and Aggarwal, 2011).

(2011), we argue for a non-linear Granger causality between returns (volatility) and volume for currency futures market. *Finally*, we provide more robustness checks for our results by showing that they are not due to divergence of opinions as measured by open interest, and that they are not sample dependent.

The results of this paper are new for the foreign currency futures markets and provide empirical support to MDH since we report a strong contemporaneous relation between returns and volume. The results of linear Granger causality indicate a strong unidirectional causality from returns to trading volume, thus rejecting SAI. The results for volatility and trading volume indicate a strong bidirectional feedback relation for all four currencies lending support to SAI. The results for non-linear Granger causality between futures returns (volatility) and trading volume indicate a strong and significant bidirectional relation which is consistent with SAI hypothesis. The findings of non-linear relation between futures returns (volatility) and trading volume are insensitive to heterogeneity of beliefs as measured by open interest and our results are not sample dependent. Therefore, we conclude that short- to medium-term movements in the currency markets are majorly influenced by microstructure and trading dynamics, not by fundamentals.

The findings of this paper are very important for the participants in the market and regulators. The participants in the market require alternatives to diversify their risk. The significant between futures returns (volatility) and trading volume implies that the current trading volume help predict the futures prices and should lead to creation of more reliable hedging strategies for investment purposes. Further, it may interest the regulators who need to decide upon the appropriateness of their policies in the currency futures market. Based on returns-volume relation, they need to set forth market restrictions such as daily price movement and position limits. Greater (less) restrictions may be required if trading volume leads to more (less) fluctuations in the futures prices.

The rest of the paper is organized as follows: section 2 provides a brief survey of literature for the linear and non-linear relation between returns (volatility) and trading volume. Section 3 discusses the data, and Section 4 provides the methodology and empirical results for the linear relation between futures returns (volatility) and trading volume. Section 5 presents the results for

the non-linear relation between futures returns (volatility) and trading volume. Section 6 explains the robustness of our empirical results and finally section 7 concludes the paper.

## **2. LITERATURE REVIEW**

### **2.1. Linear Returns (Volatility) Volume Relation**

The contemporaneous relation between returns (volatility) and trading volume has received mixed evidence. Karpoff (1987) presents an extensive review of literature in this area. One of the prominent studies in the price-volume literature is provided by Cornell (1981) who reports a positive and significant contemporaneous correlation between the daily volume changes and changes in the standard deviation of daily log price for commodity futures contracts. This relation is further supported by Malliaris and Urrutia (1998) for agricultural futures contracts and Kocagil and Shachmurove (1998) in the US futures markets. Lee and Rui (2002) report a strong bidirectional relation between volatility and trading volume for the US, UK and Japanese stocks. Wang and Cheng (2004) find that stocks with high (low) volumes are exhibit with low (high) subsequent returns. Chuang et al. (2012) use a bivariate GJR-GARCH model and report a positive contemporaneous relation between trading volume and return volatility for ten Asian stock markets. Ciner (2002) also reports positive contemporaneous price volume relations.

However, Darrat, Rahman and Zhong (2003), and Abdullahi, Kouhy and Muhammad (2014) find no evidence for contemporaneous price-volume relation. Further, Bohl and Henke (2003) don't confirm MDH in 20 Polish stocks. Luu and Martens (2003) find that the daily squared return confirm the presence of MDH; however, the results based on sum of intraday squared returns reject MDH. Chen and Daigler (2008) find a support for MDH for the index, agricultural and currency futures contracts. Although the relation between price and volume in equity markets has been explored to a large extent, currency markets have received very little attention (Mougoué and Aggarwal, 2011). However, in recent times, some papers have started examining the price-volume relation in this area as well. For instance, see, Bauwens et al. (2005), and Chang (2005). In contrast, Mougoué and Aggarwal (2011) find no evidence MDH for the US dollar, the British pound and the Canadian dollar and the Japanese yen.

### **2.2. Non-Linear Returns (Volatility) Volume Relation**

A number of studies have examined non-linear price-volume relations in exchange rates. For instance, Hsieh (1989) shows that exchange rates cannot be described by linear dependence and

finds a strong evidence of the non-linear dependence. Michael, Nobay and Peel (1997) argue that the non-linear relations towards purchasing power parity arise due to presence of transactions costs. The failure of structural models to explain exchange rate movements may be due to non-linearities present in the currency rates data (Engel and Hamilton, 1990)<sup>4</sup>. Fujihara and Mougoué (1997) report significant non-linear relation between prices and volume in petroleum futures markets even after adjusting for volatility effects using GARCH filtered returns. However, Ciner (2002) finds no evidence of non-linear relation between price and volume in Tokyo commodity futures market. Further, Mougoué and Aggarwal (2011) suggest that there is no reason to believe a linear price-volume relation. They find a strong evidence of linear and non-linear bidirectional Granger causality between returns volatility and trading volume for each of the currency.

### 3. DATA DESCRIPTION

The data for all the currency futures series has been taken from NSE which represent the daily settlement prices along with trading volume. The first trade in the currency futures in India started on August 29, 2008 when National Stock Exchange (NSE) introduced the USD-INR currency pair. Therefore, for USD-INR, the data ranges from August 29, 2008 to December 29, 2014 with a total of 1518 daily observations. Riding on the huge success of the USD-INR futures trading, later, in 2010, NSE launched the contracts for other currency pairs (EUR-INR, GBP-INR and JPY-INR) as well. Therefore, the range of data for all other currencies is less (from February 1, 2010 to December 29, 2014), that is, a total of 1188 daily observations. The exchange rates are denominated as Indian rupees per unit of foreign currency except for Japanese yen which is represented as Indian rupees per 100 units of Japanese yen.

The price series has been converted in log returns as  $R_t = \ln(p_t/p_{t-1}) \times 100$ , where  $R_t$  is the futures return,  $p_t$  is the current futures price and  $p_{t-1}$  is the futures prices one period before. A large body of literature has reported a strong evidence of both linear and non-linear trends in the trading volume series (e.g., see, Gallant et al., 1992; Lee and Rui, 2002; Ciner, 2002 and Gebka

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<sup>4</sup> Other studies which report non-linear causality between returns and trading volume are: Hiemstra and Jones (1994), Fujihara and Mougoué (1997), Moosa and Silvapulle (2000), Ciner (2002), Mougoué and Aggarwal (2011) and Gebka and Wohar (2013).

and Wohar, 2013). To allow for a non-linear trend in the trading volume, we include a quadratic trend term in the following regression as:

$$Vol_t = c_0 + c_1 t + c_2 t^2 + e_t \quad (1)$$

where  $Vol_t$  is the log of trading volume at time  $t$  for each currency. The results of equation (1) are found to be very significant and trading volume series for each currency exhibits strong quadratic time trend<sup>5</sup>. Therefore, we use detrended trading volume ( $V_t$ ) measured as the residuals of regression equation (1) for the further analyses. Table I presents the descriptive statistics for the daily currency futures returns and detrended trading volume.

**TABLE I**  
Descriptive Statistics for Currency Futures Returns and Detrended Trading Volume

|      | Mean   | SD     | Skewness | Kurtosis | J-B       | ADF       | PP        |
|------|--------|--------|----------|----------|-----------|-----------|-----------|
| RUSD | 0.0243 | 0.0158 | 0.1591   | 6.2341   | 667.95*** | -36.59*** | -36.68*** |
| VUSD | 0.0000 | 0.0081 | -0.7919  | 6.7564   | 1051.2*** | -28.48*** | -28.46*** |
| RGBP | 0.0249 | 0.0193 | 0.2197   | 4.9551   | 253.98*** | -31.10*** | -31.11*** |
| VGBP | 0.0000 | 0.0019 | -0.3814  | 2.7398   | 41.085*** | -29.64*** | -29.38*** |
| REUR | 0.0156 | 0.0160 | 0.2968   | 5.3604   | 374.68*** | -30.18*** | -30.04*** |
| VEUR | 0.0000 | 0.0073 | -0.3784  | 3.0278   | 36.275*** | -24.03*** | -23.72*** |
| RJPY | 0.0022 | 0.0184 | 0.1143   | 5.4564   | 384.95*** | -33.42*** | -33.41*** |
| VJPY | 0.0000 | 0.0065 | -0.5877  | 6.8380   | 1019.1*** | -23.44*** | -23.04*** |

**Note:** SD is the standard deviation; J-B is the Jarque-Bera test statistics for normality that is approximately chi-squared distributed with two degrees of freedom (critical value at the 5% level  $\chi^2_{0.95, 2} = 5.99$ ). ADF is the Augmented Dickey Fuller and PP is Phillips-Perron test-statistics to test the stationarity of the series. Sample Period is from August 29, 2008 to December 29, 2014 for USD-INR; and February 1, 2010 to December 29, 2014 for all other currencies. The symbols \*\*\* and \* indicate that the parameter is significant at 1% and 10% level respectively.

The results show that currency futures returns have significant skewness and excess kurtosis. The distribution of the daily returns is positively skewed and has excess kurtosis compared to the normal distribution. Further, as expected, given that detrended trading volume is just the residuals from equation (1); it has a mean of zero for all currency futures. The standard deviations of the detrended trading volume are less than that of the futures returns. The data for volume, however, are negatively skewed and display significantly higher values for kurtosis similar to returns. Fujihara and Mougoue (1997) suggest that the excess kurtosis is displayed due to a high time-varying variance. The Jarque-Bera statistic for normal distribution shows that the

<sup>5</sup> The results for equation (1) have not been presented in the paper to conserve space. However, they are available upon request.

normality hypothesis can be rejected at 1% level of significance for both detrended trading volume and futures returns for each currency.

Further, Table I also shows the results for Augmented Dicky Fuller (ADF) tests for detrended trading volume as well as futures returns for all currencies. The null hypothesis under ADF test is that a series contains a unit root, that is, it is non-stationary. As shown in the table, the futures returns and trading volume are found to be stationary and the null of unit root is rejected at 1% level of significance for all currencies. The table also reports the results for Phillips-Perron (PP) test which confirms the ADF test for all currencies. The PP test produces more robust results even in presence of high serial correlation and heteroskedasticity (Lee and Rui, 2002).

#### 4. METHODOLOGY AND EMPIRICAL RESULTS

##### 4.1. Contemporaneous Relation between Returns and Volume

We investigate the price-volume relation using an instrumental estimator as GMM to avoid the simultaneous bias problems. The GMM is a regression framework that is robust to heteroskedasticity and autocorrelation in the residuals. GMM chooses the parameter estimates in a way such that the theoretical relation is satisfied as closely as possible using its sample counterpart (known as instrumental variables) and the estimates are chosen to minimize the weighted distance between the theoretical and actual values (Foster, 1995; Lee and Rui, 2002; Mougoué and Aggarwal, 2011). GMM is a robust estimator unlike maximum likelihood estimation, in that; it does not require information of the exact distribution of the disturbances.

The model can be estimated using the following structural equation:

$$\begin{aligned} R_t &= a_0 + a_1 V_t + a_2 V_{t-1} + a_3 R_{t-1} + \varepsilon_t \\ V_t &= b_0 + b_1 R_t + b_2 V_{t-1} + b_3 R_{t-1} + \mu_t \end{aligned} \quad (2)$$

where  $R_t$  and  $V_t$  represents the daily currency futures returns and detrended trading volume respectively. The contemporaneous relation between returns and volume can be determined by examining the significance of the coefficients  $a_1$  and  $b_1$  in equation (2). If these two coefficients are found to be significant, we can establish a contemporaneous relation between the futures returns and trading volume. Table II presents the results of equation (2). The main interest of this paper for equation (2) is in the sign and significance of the coefficients  $a_1$  and  $b_1$ .

**TABLE II**  
Contemporaneous relationship between returns and detrended volume

$$R_t = a_0 + a_1 V_t + a_2 V_{t-1} + a_3 R_{t-1} + \varepsilon_t$$

$$V_t = b_0 + b_1 R_t + b_2 V_{t-1} + b_3 R_{t-1} + \mu_t$$

|                | USD       | <i>t</i> -stat | EUR      | <i>t</i> -stat | GBP      | <i>t</i> -stat | JPY       | <i>t</i> -stat |
|----------------|-----------|----------------|----------|----------------|----------|----------------|-----------|----------------|
| $a_0$          | 0.004     | (1.216)        | 0.002    | (0.750)        | 0.000    | (0.319)        | 0.003     | (1.244)        |
| $a_1$          | 0.001***  | (2.704)        | 0.001*** | (2.779)        | 0.003**  | (2.365)        | 0.001**   | (2.380)        |
| $a_2$          | -0.002*** | (-3.096)       | -0.001** | (-2.142)       | -0.002   | (-1.408)       | -0.001*** | (-2.847)       |
| $a_3$          | 0.058**   | (2.046)        | 0.130*** | (4.116)        | 0.101*** | (3.279)        | 0.028*    | (1.793)        |
| <i>J</i> -stat | 2.57E-30  |                | 5.1E-27  |                | 8.88E-29 |                | 2.4E-27   |                |
| $b_0$          | 1.386***  | (5.299)        | 1.900*** | (8.067)        | 1.261*** | (7.600)        | 1.545***  | (6.700)        |
| $b_1$          | 6.629***  | (2.911)        | 3.687**  | (2.357)        | 1.103*** | (3.185)        | 7.310**   | (2.266)        |
| $b_2$          | 0.897***  | (46.76)        | 0.814*** | (35.71)        | 0.866*** | (49.25)        | 0.824***  | (32.06)        |
| $b_3$          | 1.972     | (1.528)        | 0.454    | (0.248)        | 4.245*   | (1.734)        | 0.891     | (0.365)        |
| <i>J</i> -stat | 3.77E-26  |                | 1.81E-25 |                | 3.75E-26 |                | 1.23E-27  |                |

**Note:** The table shows the contemporaneous relationship between currency futures returns and detrended trading volume. Sample Period is from August 29, 2008 to December 29, 2014 for USD–INR; February 1, 2010 to December 29, 2014 for all other currencies. The symbols \*\*\*, \*\* and \* indicate that the parameter is significant at 1%, 5% and 10% level respectively. The results shown are after incorporating the EGARCH effect.

The results indicate a strong contemporaneous relation between currency futures returns and the trading volume. For example, for a regression of futures returns on trading volume, the coefficient  $a_1$  is found to be 0.001 for USD, 0.001 for EUR, 0.003 for GBP and 0.001 for JPY and is highly significant. Similarly, for the regression of trading volume on the futures returns, the coefficient  $b_1$  is also found to be highly significant being 6.629, 3.687, 1.103 and 7.310 for USD, EUR, GBP and JPY respectively. The *J*-statistic also suggests that the data fits the model for all the currencies. As both the coefficients  $a_1$  and  $b_1$  are reported to be highly significant, we conclude that there exists a high degree of positive contemporaneous relation between futures returns and trading volume in all the four currencies and provide empirical support to MDH. These results confirm the previous findings (e.g., see, Kocagil and Shachmurove, 1998; Ciner, 2002; Lee and Rui, 2002; and Chuang et al., 2012).

The results in Table II are presented after controlling for heteroskedasticity using the exponential generalized autoregressive conditional heteroskedasticity (EGARCH) model (explained later in the text). The results indicate that all the coefficients of EGARCH model are positive and significant at least at 5% level of significance for all the currencies. The LR statistic is also found to be very large which indicates that EGARCH is an ideal representation of the daily behavior of

currency futures returns; and successfully captures the temporal variations of returns.<sup>6</sup> Therefore, positive contemporaneous relation between the futures returns and the trading volume persists even after controlling for heteroskedasticity.

Following Bessembinder and Seguin (1992), Lee and Rui (2002), Chen (2012) and Abdullahi et al. (2014), we provide robustness tests for the results using alternative measure of trading volume such as unexpected volume. We incorporate VAR which consists of each currency's futures returns, conditional volatility and trading volume and innovations in the trading volume are used as unexpected trading volume. Specifically, we regress trading volume on a constant, past futures returns, past conditional volatilities and past trading volume; the residuals of the regression are used as unexpected volume. The tests of ADF and PP suggest that unexpected trading volumes are found to be stationary. The results obtained using unexpected volume are significantly similar to those presented in Table II and show no substantial change in the interpretation<sup>7</sup>.

#### 4.2. Linear Causal Relation between Returns (Volatility) and Volume

Granger (1969) introduced a causality test through which this paper examines whether currency futures returns (or volatility) and trading volume have forecasting ability for each other. We estimate the following bivariate vector autoregression (VAR) model:

$$\begin{aligned} X_t &= \alpha_1 + \sum_{i=1}^n \beta_i V_{t-i} + \sum_{j=1}^m \delta_j X_{t-j} + \pi_{X,t} \\ V_t &= \alpha_2 + \sum_{i=1}^n \vartheta_i X_{t-i} + \sum_{j=1}^m \theta_j V_{t-j} + \pi_{V,t} \end{aligned} \quad (3)$$

where,  $X_t$  is the proxy for futures returns, ( $R_t$ ) and conditional volatility, ( $h_t$ ) at time  $t$ ;  $V_t$  corresponds to the trading volume (detrended and unexpected) at time  $t$ . The Granger causality method tests the hypothesis  $H_0: \beta_i = 0$  which means  $V_t$  does not linear Granger cause  $X_t$ . Similarly, another hypothesis is  $H_0: \vartheta_i = 0$  according to which  $X_t$  does not linear Granger cause  $V_t$ .

We have conducted a thorough analysis for all specifications to find the best estimate of conditional volatility in equation (3). This step is highly important since if the model is misspecified, the measure of variance might be problematic, unnecessarily leading to invalid

<sup>6</sup> We have not provided the EGARCH results in the text to save space; however, they are available upon request.

<sup>7</sup> These results have not been presented to save space; however, the same are available upon request.

inferences. Moreover, focusing on the one period jump in the volatility, GARCH may not be the best approach to estimate volatility. Therefore, after a close examination of probable volatility models such as GARCH, IGARCH and TARCH; we finally arrived at the conclusion that EGARCH model is the best model as far as goodness of fit to measure volatility is concerned. The model is represented as:

$$R_t = \varphi_0 + \sum_{i=1}^{\lambda} \varphi_i R_{t-1} + \rho \sqrt{h_t} + \eta_t$$

$$\eta_t | (\eta_{t-1}, \eta_{t-2}, \eta_{t-3}, \dots) \approx GED(0, h_t), \quad (4)$$

$$h_t = \exp \left( \gamma \ln(h_{t-1}) + \phi \left[ \tau \left( \frac{|\eta_{t-1}|}{h_{t-1}} - E \left( \frac{|\eta_{t-1}|}{h_{t-1}} \right) \right) + \omega \left( \frac{|\eta_{t-1}|}{h_{t-1}} \right) \right] \right)$$

where,  $\varphi_{i,s}$ ,  $\rho$ ,  $\gamma$ ,  $\phi$ ,  $\tau$ , and  $\omega$  are the parameter coefficients to be estimated and GED is the general error distribution. We use EGARCH model to estimate the conditional volatility of daily returns because we find that the currency futures returns' errors distribution do not show a constant variance. In addition, EGARCH model avoid any possible misspecification bias since it does not impose any positive restriction on the parameters.

According to the Granger causality hypothesis, if lagged values of trading volume significantly explain currency futures returns (volatility), the paper concludes  $V_t$  Granger-causes  $X_t$  and there exists unidirectional causality from trading volume to futures returns (volatility). Similarly, if lagged values of currency futures returns (volatility) are significant in explaining the trading volume, a unidirectional causality from futures returns (volatility) to trading volume is established. Moreover, when both  $\beta_i$  and  $\vartheta_i$  are significantly different from zero, a bidirectional causality or a feedback relation between trading volume and returns (volatility) is confirmed. The standard joint  $F$ -statistic is used to test the Granger causality in a VAR system (see eg., Lee and Rui, 2002 and Chen, 2012). Table III presents the results of equation (3) for linear Granger causality between futures returns, conditional volatility and trading volume (both detrended and unexpected).

The results indicate that trading volume does not Granger-cause futures returns for any of the currencies even at 10% level of significance (except for JPY, specifically unexpected volume). This finding implies that though there is positive contemporaneous relation between trading volume and futures returns, trading volume does not possess significant predictive power for the

currency futures returns when both current and past returns are present. Moreover, the results are insensitive to using unexpected volume in place of detrended volume. These results are similar to Gallant et al. (1992) who find no causal relation from trading volume to stock returns, and Lee and Rui (2002), who report a unidirectional causality from returns to volumes of US, UK and Japanese stocks. The results of unidirectional causality from futures returns to trading volume contradict the SAI theory of Copeland (1976) and Mougoué and Aggarwal, (2011).

**TABLE III**  
Causal relationship between futures returns (volatility) and trading volume

| Null Hypothesis                 | Detrended Volume |         | Unexpected Volume |         |
|---------------------------------|------------------|---------|-------------------|---------|
| $VUSD \xrightarrow{dnGC} RUSD$  | 1.635            | (0.195) | 1.134             | (0.322) |
| $RUSD \xrightarrow{dnGC} VUSD$  | 4.510**          | (0.011) | 3.540**           | (0.029) |
| $VUSD \xrightarrow{dnGC} CVUSD$ | 7.381***         | (0.001) | 7.755***          | (0.001) |
| $CVUSD \xrightarrow{dnGC} VUSD$ | 4.925***         | (0.007) | 3.234**           | (0.040) |
| $VEUR \xrightarrow{dnGC} REUR$  | 0.664            | (0.515) | 0.793             | (0.453) |
| $REUR \xrightarrow{dnGC} VEUR$  | 4.283**          | (0.014) | 3.646**           | (0.026) |
| $VEUR \xrightarrow{dnGC} CVEUR$ | 3.527**          | (0.028) | 5.013***          | (0.006) |
| $CVEUR \xrightarrow{dnGC} VEUR$ | 2.832*           | (0.089) | 3.421**           | (0.033) |
| $VGBP \xrightarrow{dnGC} RGBP$  | 1.325            | (0.194) | 1.592             | (0.204) |
| $RGBP \xrightarrow{dnGC} VGBP$  | 7.034***         | (0.001) | 6.393***          | (0.001) |
| $VGBP \xrightarrow{dnGC} CVGBP$ | 9.850***         | (0.022) | 11.49***          | (0.000) |
| $CVGBP \xrightarrow{dnGC} VGBP$ | 3.367**          | (0.035) | 4.608***          | (0.010) |
| $VJPY \xrightarrow{dnGC} RJPY$  | 1.597            | (0.203) | 2.497*            | (0.083) |
| $RJPY \xrightarrow{dnGC} VJPY$  | 3.453**          | (0.032) | 5.163***          | (0.006) |
| $VJPY \xrightarrow{dnGC} CVJPY$ | 4.903***         | (0.008) | 6.088***          | (0.002) |
| $CVJPY \xrightarrow{dnGC} VJPY$ | 7.152***         | (0.001) | 10.11***          | (0.000) |

**Note:** The table shows the Causal relationship between currency futures returns and trading volume.  $A \xrightarrow{dnGC} B$  denotes the null that A does not granger cause B. V, R and CV stand for trading volume, futures returns and conditional volatility respectively. Sample Period is from August 29, 2008 to December 29, 2014 for USD–INR; February 1, 2010 to December 29, 2014 for all other currencies. The symbols \*\*\*, \*\* and \* indicate that the parameter is significant at 1%, 5% and 10% level respectively.

When the causal relation between futures returns volatility and trading volume is analyzed, a strong bidirectional relation between the two variables is found for each of the currencies. Results indicate that for all four currencies, the null hypothesis of linear Granger non-causality from trading volume to volatility is strongly rejected, implying that past volume does have predictive ability for volatility. Similarly, Granger non-causality from volatility to trading volume is also firmly rejected for all the four currencies. Further, the results for equation (3) to test the Granger-causality between futures returns volatility and trading volume were found to be insignificantly different even up to five lags. The bidirectional relation between volatility and volume suggest that the SAI theory holds for these two variables. Through using unexpected component of trading volume, we establish that contemporaneous and causal relation between futures returns (volatility) and trading volume are robust across currencies.

## 5. NON-LINEAR RELATION BETWEEN RETURNS (VOLATILITY) AND VOLUME

The results on Granger-causality presented in Table III assume that the relation between futures returns (volatility) and trading volume is linear. Therefore, equation (3) is not able to capture any non-linear relation between returns and volume. As a result of conducting some initial non-linearity tests, we find that the residuals from equation (3) exhibit significant heteroskedasticity<sup>8</sup> and non-linearity similar to Mougoué and Aggarwal (2011). Given the evidence of non-linear dependency between all the univariate series, we analyze the application of Baek and Brock (1992) model to the residuals of equation (3) to test for non-linear Granger causality between returns (volatility) and trading volume.

We test for non-linear Granger causality between returns (volatility) and trading volume based on a modified version of the Baek and Brock (1992) nonparametric technique developed by Hiemstra and Jones (1994). For further illustration of the test, we consider the following two time series  $X_t$  [(proxy for futures returns,  $(R_t)$  and conditional volatility,  $(h_t)$ ] and  $V_t$  (trading volume) which are strictly stationary and weakly dependent. Assume that  $l$ -length lead vector of  $X_t$  is shown by  $X_t^l$ ,  $k$ -length and  $q$ -length lag vectors of  $X_t$  and  $V_t$ , respectively, by  $X_{t-k}^k$  and  $V_{t-q}^q$ , that is,

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<sup>8</sup> We conduct  $Q^2$ -statistics for the non-linear dependency between returns (volatility) and volume. The null hypothesis of the  $Q^2$ -statistic is that there is no serial correlation in squared residuals. The results indicate that the residuals for volume and returns (and volatility) show significant conditional heteroskedasticity. The results are available upon request.

$$\begin{aligned}
X_t^l &\equiv (X_t, X_{t+1}, \dots, X_{t+l-1}), & l = 1, 2, \dots, \quad t = 1, 2, \dots, \\
X_{t-k}^k &= (X_{t-k}, X_{t-k+1}, \dots, X_{t-1}), & k = 1, 2, \dots, \quad t = k+1, k+2, \dots, \\
V_{t-q}^q &= (V_{t-q}, V_{t-q+1}, \dots, V_{t-1}), & q = 1, 2, \dots, \quad t = n+1, n+2, \dots
\end{aligned} \tag{5}$$

For any given value of  $m, k$  and  $n \geq 1$ , and  $\epsilon > 0$ ,  $V_t$  fails to strictly Granger cause  $X_t$  if,

$$\begin{aligned}
&\Pr(\|X_t^l - X_s^l\| < \epsilon \mid \|X_{t-k}^k - X_{s-k}^k\| < \epsilon, \|V_{t-q}^q - V_{s-q}^q\| < \epsilon) \\
&= \Pr(\|X_t^l - X_s^l\| < \epsilon \mid \|X_{t-k}^k - X_{s-k}^k\| < \epsilon)
\end{aligned} \tag{6}$$

where  $\Pr(\cdot)$  stands for probability and  $\|\cdot\|$  represents the maximum norm. The  $\Pr$  on the left hand side of equation (6) stands for a conditional probability that two random  $l$ -length lead vectors of  $X_t$  are within a distance,  $\epsilon$ , of each other, with a condition that the respective  $k$ -length lag vector of  $X_t$  and  $q$ -length lag vector of  $V_t$  are also within a distance,  $\epsilon$ , of each other. Similarly, the  $\Pr$  on the right hand side of equation (6) is a conditional probability that two random  $l$ -length lead vectors of  $X_t$  are within a distance,  $\epsilon$ , of each other, assuming that their respective  $k$ -length lag vectors are also within a distance,  $\epsilon$ , of each other.

To test that  $V_t$  does not strictly Granger cause  $X_t$ , it is useful to express the conditional probabilities in terms of the corresponding ratios of joint probabilities. The strict Granger non-causality condition in equation (6) can be expressed as:

$$\frac{C1(l+k, q, \epsilon)}{C2(k, q, \epsilon)} = \frac{C3(l+k, \epsilon)}{C2(k, \epsilon)} \tag{7}$$

where,

$$\begin{aligned}
C1(l+k, q, \epsilon) &= \Pr(\|X_{t-k}^{l+k} - X_{s-k}^{l+k}\| < \epsilon, \|V_{t-q}^q - V_{s-q}^q\| < \epsilon) \\
C2(k, q, \epsilon) &= \Pr(\|X_{t-k}^k - X_{s-k}^k\| < \epsilon, \|V_{t-q}^q - V_{s-q}^q\| < \epsilon) \\
C3(l+k, \epsilon) &= \Pr(\|X_{t-k}^{l+k} - X_{s-k}^{l+k}\| < \epsilon) \\
C4(k, \epsilon) &= \Pr(\|X_{t-k}^k - X_{s-k}^k\| < \epsilon)
\end{aligned} \tag{8}$$

and  $C_i$  are the correlation-integral estimators of the joint probabilities. Assuming that  $X_t$  and  $V_t$  are strictly stationary and weakly dependent, if  $V_t$  does not Granger cause  $X_t$ , then,

$$\begin{aligned}
&\sqrt{p} \left[ \frac{C1(l+k, q, \epsilon, p)}{C2(k, q, \epsilon, p)} - \frac{C3(l+k, \epsilon, p)}{C2(k, \epsilon, p)} \right] \\
&\approx N(0, \sigma^2, (l, k, q, \epsilon))
\end{aligned} \tag{9}$$

The test statistics illustrated in equation (7) and (9) are applied to the estimated residual series from equation (3),  $\pi_{X,t}$  and  $\pi_{V,t}$ . The null hypothesis that  $V_t$  does not non-linearly strictly Granger cause  $X_t$  holds for all  $l$ ,  $k$  and  $q \geq 1$ , and for all  $e > 0$ . As the evidence of returns (volatility) and volume linear causality is significant; any linear predictive power with a VAR model as in equation (3) has been removed. Therefore, remaining predictive power of one residual series for another, if any, can be considered as a non-linear predictive power similar to Mougoué and Aggarwal (2011).

There are no methods available for selecting the optimal values for lag lengths ( $k$  and  $q$ ) and scale parameters ( $e$ ) for non-linear testing unlike linear Granger causality. Therefore, this paper follows the Mougoué and Aggarwal (2011) approach in which we rely on Monte Carlo results as in Hiemstra and Jones (1993) and set lead length  $l = 1$ , and  $k = q$  for all cases. Further, similar to Hiemstra and Jones (1993) and Ciner (2002), we use common lag lengths of 1–8, and a common scale parameter of  $e = 1.0 \times \sigma$ , where  $\sigma = 1$  denotes the standard deviation of standardized series<sup>9</sup>.

Non-linear Granger causality between futures returns and detrended trading volume using modified version of the Baek and Brock nonparametric technique are presented in Table IV. The results are strikingly different to those reported in Table III for their linear relation. Table IV shows an intriguing evidence of bidirectional non-linear relation between returns and trading volume in contrast to Table III, where we document a unidirectional Granger causality from returns to volume;. All the standardized test statistics (NORM) are significant at least at 5% level, except for EUR at lag 8 and GBP at lag 6 and 7, where non-linear Granger causality from trading volume to futures returns is not significant.

Compared to Table III, where no evidence of causality from volume to returns is observed, Table IV reveals that trading volume contains important information useful for forecasting the returns, though in a non-linear framework. Overall, Table IV shows strong evidence of bidirectional non-linear Granger-causality between futures returns and trading volume. Table V shows the non-linear Granger causality between volatility and volume.

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<sup>9</sup> We also test with the scale parameter of  $e = 0.5$  and  $e = 1.5$ . However, the results were found to be robust irrespective of the scale parameter chosen.

**TABLE IV**

Non-linear Granger causality between currency futures returns and trading volume

| $k = q$                                                        | USD    |          | EUR    |          | GBP    |          | JPY    |          |
|----------------------------------------------------------------|--------|----------|--------|----------|--------|----------|--------|----------|
|                                                                | DIFF   | NORM     | DIFF   | NORM     | DIFF   | NORM     | DIFF   | NORM     |
| Panel A: Futures returns do not Granger cause trading volume   |        |          |        |          |        |          |        |          |
| 1                                                              | 0.0024 | 9.802*** | 0.0039 | 7.229*** | 0.0012 | 8.445*** | 0.0008 | 8.675*** |
| 2                                                              | 0.0011 | 8.123*** | 0.0101 | 8.293*** | 0.0023 | 8.079*** | 0.0017 | 9.092*** |
| 3                                                              | 0.0130 | 7.283*** | 0.0027 | 8.554*** | 0.0031 | 9.382*** | 0.0035 | 7.877*** |
| 4                                                              | 0.0029 | 9.625*** | 0.0038 | 9.108*** | 0.0014 | 9.921*** | 0.0024 | 8.255*** |
| 5                                                              | 0.0047 | 8.115*** | 0.0041 | 7.378*** | 0.0037 | 7.863*** | 0.0015 | 7.433*** |
| 6                                                              | 0.0039 | 7.204*** | 0.0033 | 7.006*** | 0.0016 | 7.156*** | 0.0032 | 7.986*** |
| 7                                                              | 0.0021 | 6.996*** | 0.0026 | 8.107*** | 0.0022 | 6.891*** | 0.0006 | 8.089*** |
| 8                                                              | 0.0044 | 6.892*** | 0.0029 | 7.997*** | 0.0009 | 7.243*** | 0.0033 | 7.444*** |
| Panel B: Trading volume does not Granger cause futures returns |        |          |        |          |        |          |        |          |
| 1                                                              | 0.0004 | 5.247*** | 0.0052 | 6.652*** | 0.0040 | 5.121*** | 0.0045 | 4.988*** |
| 2                                                              | 0.0007 | 4.981*** | 0.0004 | 4.278**  | 0.0027 | 6.195*** | 0.0041 | 5.622*** |
| 3                                                              | 0.0009 | 4.108**  | 0.0011 | 4.937*** | 0.0008 | 5.946*** | 0.0036 | 5.449*** |
| 4                                                              | 0.0002 | 7.726*** | 0.0036 | 5.119*** | 0.0018 | 4.274**  | 0.0039 | 5.367*** |
| 5                                                              | 0.0005 | 7.843*** | 0.0001 | 4.0383** | 0.0025 | 4.871*** | 0.0028 | 4.891*** |
| 6                                                              | 0.0001 | 5.848*** | 0.0048 | 4.611*** | 0.0016 | 2.982*   | 0.0034 | 3.661**  |
| 7                                                              | 0.0004 | 3.289**  | 0.0003 | 4.101**  | 0.0042 | 2.164    | 0.0019 | 3.434**  |
| 8                                                              | 0.0003 | 4.719*** | 0.0019 | 2.189    | 0.0007 | 3.222**  | 0.0021 | 4.194**  |

**Note:** The table shows the non-linear Granger causality between futures returns and trading volume using modified version of the Baek and Brock (1992) nonparametric technique. Sample Period is from August 29, 2008 to December 29, 2014 for USD–INR; February 1, 2010 to December 29, 2014 for all other currencies. The lead length  $l$  is set equal to 1, and  $k = q$  for all cases and common scale parameter of  $e$  is set to equal to 1. DIFF is the difference between the two conditional probabilities and NORM is the standardized test statistic which is asymptotically distributed as  $N(0, 1)$ . The symbols \*\*\*, \*\* and \* indicate that the parameter is significant at 1%, 5% and 10% level respectively.

**TABLE V**

Non-linear Granger causality between futures returns volatility and trading volume

| $k = q$                                          | USD    |           | EUR    |           | GBP    |           | JPY    |           |
|--------------------------------------------------|--------|-----------|--------|-----------|--------|-----------|--------|-----------|
|                                                  | DIFF   | NORM      | DIFF   | NORM      | DIFF   | NORM      | DIFF   | NORM      |
| Returns volatility does not Granger cause volume |        |           |        |           |        |           |        |           |
| 1                                                | 0.0032 | 11.228*** | 0.0041 | 10.345*** | 0.0018 | 9.490***  | 0.0025 | 10.444*** |
| 2                                                | 0.0016 | 10.666*** | 0.0005 | 9.135***  | 0.0005 | 7.142***  | 0.0027 | 10.371*** |
| 3                                                | 0.0014 | 9.832***  | 0.0042 | 10.411*** | 0.0039 | 11.203*** | 0.0045 | 8.117***  |
| 4                                                | 0.0022 | 8.563***  | 0.0017 | 9.743***  | 0.0044 | 9.488***  | 0.0048 | 9.824***  |
| 5                                                | 0.0035 | 9.337***  | 0.0024 | 9.808***  | 0.0031 | 9.731***  | 0.0055 | 8.188***  |
| 6                                                | 0.0031 | 8.115***  | 0.0037 | 11.285*** | 0.0008 | 7.878***  | 0.0042 | 8.262***  |
| 7                                                | 0.0014 | 7.592***  | 0.0015 | 8.766***  | 0.0034 | 8.622***  | 0.0019 | 8.634***  |
| 8                                                | 0.0008 | 6.471***  | 0.0022 | 9.399***  | 0.0027 | 7.694***  | 0.0024 | 6.297***  |
| Volume does not Granger cause returns volatility |        |           |        |           |        |           |        |           |
| 1                                                | 0.0009 | 8.727***  | 0.0038 | 8.555***  | 0.0026 | 8.352***  | 0.0051 | 6.176***  |
| 2                                                | 0.0017 | 6.007***  | 0.0013 | 7.712***  | 0.0036 | 7.865***  | 0.0016 | 7.163***  |
| 3                                                | 0.0021 | 8.232***  | 0.0019 | 6.105***  | 0.0013 | 7.169***  | 0.0029 | 4.226**   |
| 4                                                | 0.0039 | 9.844***  | 0.0043 | 8.269***  | 0.0028 | 6.328***  | 0.0017 | 4.112**   |
| 5                                                | 0.0011 | 7.333***  | 0.0017 | 7.177***  | 0.0009 | 2.871*    | 0.0007 | 3.141**   |
| 6                                                | 0.0023 | 7.812***  | 0.0008 | 7.262***  | 0.0041 | 3.007**   | 0.0014 | 3.318**   |
| 7                                                | 0.0033 | 5.197***  | 0.0014 | 4.576***  | 0.0009 | 4.721***  | 0.0009 | 2.189     |
| 8                                                | 0.0028 | 4.166**   | 0.0006 | 3.193**   | 0.0023 | 4.161**   | 0.0038 | 4.564***  |

**Note:** The table shows the non-linear Granger causality between futures returns volatility and trading volume using modified version of the Baek and Brock (1992) nonparametric technique. Sample Period is from August 29, 2008 to December 29, 2014 for USD–INR; February 1, 2010 to December 29, 2014 for all other currencies. The lead length  $l$  is set equal to 1, and  $k = q$  for all cases and common scale parameter of  $e$  is set to equal to 1. DIFF is the difference between the two conditional probabilities and NORM is the standardized test statistic which is asymptotically distributed as  $N(0, 1)$ . The symbols \*\*\*, \*\* and \* indicate that the parameter is significant at 1%, 5% and 10% level respectively.

The results are found to be quite similar to those reported in Table III and show a strong bidirectional non-linear relation between returns volatility and trading volume. For the Granger causality from returns volatility to trading volume, the standardized test statistic (NORM) is significant at 1% level for each of the currency. Similarly, Granger causality from trading volume to returns volatility is also very strong at least at 5% level, except for GBP at lag 5 and JPY at lag 7.

The findings of non-linear relation between futures returns (volatility) and trading volume<sup>10</sup> can be explained by the concept of noise traders developed by DeLong, Shleifer, Summers and Waldmann (1990) and Chen and Daigler (2008). They suggest that due to the presence of noise traders in the market, prices have a tendency to deviate from their intrinsic value as they trade on the basis of random and erroneous expectations. Consequently, imperfections arise in the prices which lead to non-linear dynamic relation between returns and trading volume. The second explanation is derived from Mougoué and Aggarwal (2011) who state that non-linear relations between volume and volatility arise due to heterogeneity of beliefs.

## **6. ROBUSTNESS TESTS**

### **6.1. Non-linear causality using open interest filtered data**

In this section, we provide robustness results to confirm the non-linear relation between trading volume and conditional volatility. As explained in the previous section, the heterogeneity of beliefs or divergence of opinions could lead to the possible non-linear relation between volume and volatility. Kim and Verrecchia (1991) state that new information release changes investors' opinions which lead to another round of trading volume. Further, Shalen (1993), shows that divergence of opinions contributes towards positive correlations between trading volume and price changes; which in turn explains the excess trading volume in the market. More the divergence of belief more would the volume increase.

Following Mougoué and Aggarwal (2011), we use open interest as a measure of divergence of beliefs. To examine the role of divergence of opinions, we again estimate the EGARCH model

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<sup>10</sup> The results of non-linear relationship between futures returns (volatility) and trading volume were also analyzed using unexpected trading volume in place of detrended volume. These results were insignificantly different from those reported in Tables IV and V, and therefore not presented in the text.

as in equation (4) incorporating open interest (*OI*) in both mean and variance equations. We specifically estimate the following equation:

$$R_t = \varphi_0 + \sum_{i=1}^{\lambda} \varphi_i R_{t-i} + \rho\sqrt{h_t} + \psi OI + \eta_t$$

$$\eta_t | (\eta_{t-1}, \eta_{t-2}, \eta_{t-3}, \dots) \approx GED(0, h_t), \quad (10)$$

$$h_t = \exp \left( \gamma \ln(h_{t-1}) + \kappa OI + \phi \left[ \tau \left( \frac{|\eta_{t-1}|}{h_{t-1}} - E \left( \frac{|\eta_{t-1}|}{h_{t-1}} \right) \right) + \omega \left( \frac{|\eta_{t-1}|}{h_{t-1}} \right) \right] \right)$$

where all the variables carry the same meaning as explained in equation (4). We have not presented the results of equation (10) to save some space, however we do mention the major findings. The *OI* is found to be statistically significant in both mean and variance equations for all the currency futures contracts providing support to the theoretical work of Shalen (1993). We report that *OI* significantly explains both the mean and conditional variance of trading volume similar to Bessembinder et al. (1996) who find that *OI* is an important determinant of volume for both S&P 500 stocks and futures. Though, the persistence in conditional variance is somewhat reduced, however, it is still present. Therefore, we believe that the documented persistence of variance in the literature is attributable to other factors in the currency markets other than divergence of beliefs.

We apply the modified version of the Baek and Brock (1992) nonparametric technique to test for bidirectional non-linear Granger causality between futures returns volatility and trading volume. Non-linear Granger causality between conditional volatility and detrended trading volume after filtering the data for *OI* effects are presented in Table VI<sup>11</sup>. We report the similar and strong evidence of bidirectional non-linear causality between volatility and trading volume for all four currencies considered, although the significance levels have come down marginally. Our results confirm the notion that the divergence of opinions measured by *OI* explains some of the non-linear relations between trading volume and returns volatility. However, a strong bidirectional non-linear relation between volatility and volume suggest that there might be other factors that may be explaining this relationship (Mougoué and Aggarwal, 2011). The results reported in Table V and robust findings in Table VI for an emerging market are similar to those reported by studies on developed markets such as Mougoué and Aggarwal (2011).

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<sup>11</sup> The results were insensitive to using unexpected trading volume in place of detrended volume.

**TABLE VI**

Non-linear Granger causality between futures returns volatility and trading volume filtered for open interest

| $k = q$                                          | USD    |           | EUR    |           | GBP    |           | JPY    |           |
|--------------------------------------------------|--------|-----------|--------|-----------|--------|-----------|--------|-----------|
|                                                  | DIFF   | NORM      | DIFF   | NORM      | DIFF   | NORM      | DIFF   | NORM      |
| Returns volatility does not Granger cause volume |        |           |        |           |        |           |        |           |
| 1                                                | 0.0012 | 10.183*** | 0.0004 | 9.165***  | 0.0008 | 9.708***  | 0.0014 | 9.731***  |
| 2                                                | 0.0010 | 8.248***  | 0.0022 | 9.052***  | 0.0033 | 6.883***  | 0.0008 | 10.710*** |
| 3                                                | 0.0004 | 9.204***  | 0.0051 | 9.217***  | 0.0020 | 9.148***  | 0.0015 | 6.876***  |
| 4                                                | 0.0026 | 9.282***  | 0.0039 | 7.107***  | 0.0017 | 9.201***  | 0.0061 | 8.405***  |
| 5                                                | 0.0019 | 7.174***  | 0.0016 | 8.105***  | 0.0006 | 10.317*** | 0.0047 | 8.606***  |
| 6                                                | 0.0009 | 7.905***  | 0.0030 | 10.209*** | 0.0015 | 7.166***  | 0.0036 | 6.007***  |
| 7                                                | 0.0018 | 5.026***  | 0.0021 | 9.163***  | 0.0026 | 6.561***  | 0.0030 | 7.518***  |
| 8                                                | 0.0021 | 6.810***  | 0.0029 | 8.731***  | 0.0011 | 6.492***  | 0.0019 | 5.390***  |
| Volume does not Granger cause returns volatility |        |           |        |           |        |           |        |           |
| 1                                                | 0.0016 | 6.156***  | 0.0029 | 7.150***  | 0.0010 | 7.129***  | 0.0060 | 7.912***  |
| 2                                                | 0.0019 | 5.896***  | 0.0011 | 5.374***  | 0.0051 | 5.252***  | 0.0028 | 6.453***  |
| 3                                                | 0.0038 | 8.904***  | 0.0022 | 5.564***  | 0.0019 | 6.482***  | 0.0047 | 5.109**   |
| 4                                                | 0.0044 | 7.267***  | 0.0009 | 6.190***  | 0.0033 | 7.673***  | 0.0032 | 3.488**   |
| 5                                                | 0.0027 | 6.461***  | 0.0033 | 7.804***  | 0.0048 | 4.649***  | 0.0015 | 2.817*    |
| 6                                                | 0.0017 | 7.280***  | 0.0017 | 6.493***  | 0.0037 | 3.261**   | 0.0033 | 5.461***  |
| 7                                                | 0.0008 | 4.554***  | 0.0036 | 4.168**   | 0.0011 | 3.898**   | 0.0017 | 3.228**   |
| 8                                                | 0.0013 | 4.052**   | 0.0021 | 2.884*    | 0.0039 | 3.781**   | 0.0012 | 3.108**   |

**Note:** The table shows the non-linear Granger causality between futures returns volatility and trading volume using modified version of the Baek and Brock (1992) nonparametric technique. Sample Period is from August 29, 2008 to December 29, 2014 for USD–INR; February 1, 2010 to December 29, 2014 for all other currencies. The lead length  $l$  is set equal to 1, and  $k = q$  for all cases and common scale parameter of  $e$  is set to equal to 1. DIFF is the difference between the two conditional probabilities and NORM is the standardized test statistic which is asymptotically distributed as  $N(0, 1)$ . The symbols \*\*\*, \*\* and \* indicate that the parameter is significant at 1%, 5% and 10% level respectively.

## 6.2. Sub-period analysis

In this subsection, we investigate the robustness of our results considering some major events happening in the Indian currency futures market which could have induced structural changes in the data. For instance, on December 15, 2011, the RBI prohibited the rebooking of forward contracts on the Indian rupee<sup>12</sup>. Further, on July 8, 2013, RBI disallowed the proprietary trading by Category-I banks and Securities and Exchange Board of India (SEBI) decreased the position limits and raised the margins on currency futures contracts traded on exchanges<sup>13</sup>. To examine the impact of these major events on our results, we classify the data into three sub-periods. We break the data first on December 15, 2011 and then on July 8, 2013 for all futures contracts. We conduct the entire analysis for the linear and non-linear relationship between futures returns (volatility) and trading volume (detrended and unexpected), and find the new results to be qualitatively similar to those reported in Tables III, IV and V.

To save space, we do not document the results of sub-period analysis in the text. However, we do point out here that our sub-periods results are supportive of unidirectional linear relation between returns and volume, and bidirectional linear relation between volatility and returns. Further, the results for non-linear causality also confirm the presence of significant bidirectional non-linear relation between futures returns (volatility) and volume.

## 7. CONCLUSION

This article investigates the linear and non-linear relation between futures returns (volatility) and trading volume by using the daily data for four currency futures contracts USD-INR, EUR-INR, GBP-INR, and JPY-INR. To the best of our knowledge, this is the first study to examine the relation between futures returns (volatility) and trading volume using the data on currency futures contracts from NSE where the trading was introduced only in August 2008.

The results of GMM indicate a strong contemporaneous relation between futures returns and trading volume lending a strong support to MDH. The results for Granger relation show strong linear unidirectional causality from futures returns to trading volume. The trading volume, however, does not contain any information to predict the future pattern of the returns contradicting the SAI theory. In contrast, the results for the Granger causality between volatility

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<sup>12</sup> <https://rbidocs.rbi.org.in/rdocs/notification/PDFs/EAP58151211FL.pdf>

<sup>13</sup> <https://rbidocs.rbi.org.in/rdocs/notification/PDFs/07APDCF080713.pdf>

and trading volume reveal a strong and positive bidirectional relation between the two variables which are insensitive to the number of lags analyzed supporting the SAI hypothesis. The results based on the correlation integrals show a strong evidence of positive bidirectional non-linear relation between futures returns (volatility) and trading volume supporting the SAI hypothesis. The findings of non-linear relation between futures returns (volatility) and trading volume can be explained by the argument of noise traders and heterogeneity of beliefs. Our results are insensitive to the use of alternate measure of trading volume such as detrended and unexpected trading volume. Further, our results are robust to divergence of opinions as measured by open interest and are not sample dependent.

The findings of this paper contain important information for the regulators who need to decide upon the aptness of their strategies in the currency futures market. They would put more restrictions if increase in trading volume causes more fluctuations in the futures prices and vice-versa. Overall, the relation between currency futures returns and trading volume in the Indian market shows dynamics that are similar to more established currency futures markets. Moreover, the short- to medium-term movements in the currency markets are dominated by market microstructural and trading issues rather than macro and fundamental variables. Nonetheless, this is a relatively new and developing market; we recommend further research on dynamics of currency futures returns-volume relation when more data is available.

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